Dual Prototype Evolving for Test-Time Generalization of Vision-Language Models

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Project Page Paper

NEURAL INFORMATION PROCESSING SYSTEMS

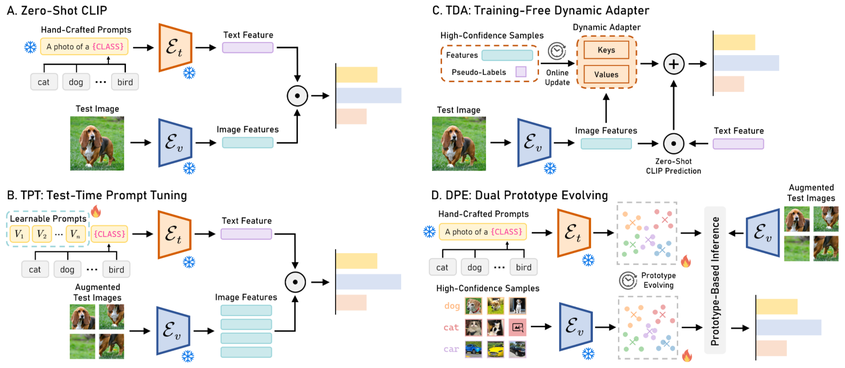
Introduction

- **Background**: Current VLMs excel in zero-shot general-purpose classifications, but struggle in transferring to specific downstream tasks, primarily due to distribution shifts between pre-training and task-specific data.
- **Objective**: In this work, we propose a novel method to leverage ***only the unlabeled test data*** to improve the performance of VLMs to out-of-distribution domains.
- **Limitations of existing works**:
  - I. We expect that with more seen samples, the performance should improve as task-specific knowledge accumulates. However, existing methods treat each test sample as ***independent*** and need to ***restart from the original model*** for each sample.
  - II. Effective adaptation of VLMs benefits from leveraging knowledge ***from both textual and visual modalities***. However, previous works only capture domain-specific knowledge from a single modality.
- **Our Approach**: To this end, we propose **Dual Prototype Evolving (DPE)**, a novel test-time VLM adaptation approach that effectively ***accumulates*** task-specific knowledge from ***multi-modalities***.

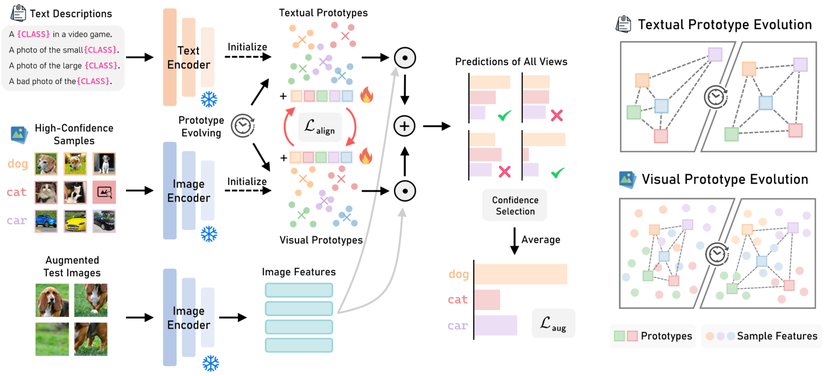
Method Overview

We introduce **DPE** to enhance CLIP's **zero-shot generalization** capabilities across diverse data distributions **during test time**.

- We introduce prototypes (*i.e.* the average representation of a target class) from **both textual and visual modalities** to refine class representations.
- For each test sample, we optimize both prototypes using learnable residual parameters with **alignment** loss and self-entropy loss.
- These prototypes **progressively evolve** as more unlabeled samples arrive, thus refining prototypes and target classes.



DPE: Dual Prototype Evolving



★ **Textual Prototype Evolution.** We use multiple text descriptions and compute the prototypes in the embedding space. To improve the quality of the textual prototypes over time, we design them to be updated online through a ***cumulative average*** with each individual sample in the test stream.

★ **Visual Prototype Evolution.** We design a ***priority queue strategy*** to store the top-*M* image features for each class based on ***predicted pseudo-labels*** and the ***corresponding self-entropy***. The priority queues are updated iteratively during testing by replacing low-confidence image features with high-confidence ones.

★ **Prototype-Based Inference.** Based on our two sets of multi-modal prototypes, the final prediction is given by

$$f_v = \mathcal{E}_v(X), \quad \mathbb{P}_{\text{Proto}}(y = y_c|X) = \frac{\exp\left(\left(f_v^\top \mathbf{t}_c + \mathcal{A}(f_v^\top \mathbf{v}_c)\right)/t\right)}{\sum_{c'} \exp\left(\left(f_v^\top \mathbf{t}_{c'} + \mathcal{A}(f_v^\top \mathbf{v}_{c'})\right)/t\right)}$$

★ **Prototype Residual Learning**. To further improve the zero-shot generalizability of our method, we introduce prototype residual learning, which optimizes multi-modal prototypes for each test sample. Apart from the self-entropy loss, we introduce a ***contrastive alignment loss*** to promote consistent multi-modal representations

Takeaways

- ✓ We propose **Dual Prototype Evolving (DPE)**, a novel test-time adaptation method for VLMs that ***progressively*** captures more accurate ***multi-modal*** representations for target classes during test time.
- ✓ To promote **consistent** multi-modal representations, we introduce and optimize **learnable residuals** for each test sample to align the prototypes across modalities. DPE **does not need** to back-propagate gradients through the textual encoder of CLIP.
- ✓ Experimental evaluations demonstrate that our DPE consistently outperforms current state-of-the-art methods across **15 diverse datasets** while maintaining **competitive computational efficiency**.

Experiments

We evaluate our proposed method on ***robustness to natural distribution shifts*** and ***cross-datasets generalization*** across 15 various dataset.

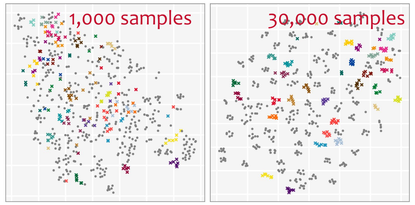
Performance comparisons on **robustness to natural distribution shifts**

Method	ImageNet	ImageNet-A	ImageNet-V2	ImageNet-R	ImageNet-S	Average	OOD Average
CLIP-ResNet-50 [44]	58.16	21.83	51.41	56.15	33.37	44.18	40.69
TPT [52]	60.74	26.67	54.70	59.11	35.09	47.26	43.89
DiffTPT [11]	60.80	**31.06**	55.80	58.80	37.10	48.71	45.69
TDA [24]	61.35	30.29	55.54	62.58	38.12	49.58	46.63
Ours	**63.41**	30.15	**56.72**	**63.72**	**40.03**	**50.81**	**47.66**
CLIP-ViT-B/16 [44]	66.73	47.87	60.86	73.98	46.09	59.11	57.20
TPT [52]	68.98	54.77	63.45	77.06	47.94	62.44	60.81
DiffTPT [11]	70.30	55.68	65.10	75.00	46.80	62.28	60.52
TDA [24]	69.51	**60.11**	64.67	80.24	50.54	65.01	63.89
Ours	**71.91**	59.63	**65.44**	**80.40**	**52.26**	**65.93**	**64.43**

Performance comparisons on **cross-datasets generalization**

Method	Aircraft	Caltech	Cars	DTD	EuroSAT	Flower	Food101	Pets	SUN397	UCF101	Average
CLIP-ResNet-50	15.66	85.88	55.70	40.37	23.69	61.75	73.97	83.57	58.80	58.84	55.82
TPT [52]	17.58	87.02	58.46	40.84	28.33	62.69	74.88	84.49	61.46	60.82	57.66
DiffTPT [11]	17.60	86.89	**60.71**	40.72	41.04	63.53	**79.21**	83.40	62.72	62.67	59.85
TDA [24]	17.61	89.70	57.78	43.74	**42.11**	**68.74**	77.75	86.18	62.53	**64.18**	61.03
Ours	**19.80**	**90.83**	59.26	**50.18**	41.67	67.60	77.83	85.97	**64.23**	61.98	**61.93**
CLIP-ViT-B/16	23.67	93.35	65.48	44.27	42.01	67.44	83.65	88.25	62.59	65.13	63.58
TPT [52]	24.78	94.16	66.87	47.75	42.44	68.98	84.67	87.79	65.50	68.04	65.10
DiffTPT [11]	25.60	92.49	67.01	47.00	43.13	70.10	**87.23**	88.22	65.74	62.67	65.47
TDA [24]	23.91	94.24	67.28	47.40	**58.00**	71.42	86.14	88.63	67.62	**70.66**	67.53
Ours	**28.95**	**94.81**	**67.31**	**54.20**	55.79	**75.07**	86.17	**91.14**	**70.07**	70.44	**69.40**

t-SNE visualizations of the priority queue



- Our priority queue strategy effectively **accumulates** high-confidence samples, **progressively refining** the visual prototypes over time.

Efficiency comparison

Method	Testing Time	Accuracy	Gain
CLIP [45]	9 min	59.81	-
TPT [53]	9 h 15 min	60.74	+0.93
DiffTPT [12]	> 20 h	60.80	+0.99
TDA [26]	1 h 5 min	61.35	+1.54
TPS [55]	55 min	61.47	+1.66
DPE (Ours)	1 h 50 min	**63.41**	**+3.60**

- Our proposed DPE not only achieves superior performance but also demonstrates over **5× and 10× greater** test-time efficiency **compared to TPT and DiffTPT**, respectively.

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